LESSON 10 · MISSION GUIDE

The Council of Guilds

groupby is the PivotTable, merge is XLOOKUP. Together they are most of the pay gap report.

From	**Queen Aldara of Thornmere**
To	**Jen, Keeper of the Ledger**
By	**raven, at dusk on the last day of the week**

Keeper, the Council of Guilds sits in three weeks and wants the pay gap "by guild, by town, by grade, with the bonus matter". Last year Lady Siv built eleven PivotTables by hand and one pointed at the wrong tab. We found out in the Council chamber, in front of eight Guildmasters.

Orrin says pandas does a PivotTable in one line (`groupby`) and XLOOKUP in one line (`merge`), so the eleven tables would be eleven lines. If that is true, build them. If not, tell Orrin before the Council sits, not during. No pressure. (Some pressure.) — A.

What you'll be able to do after today

- [ ] Count things by category with `value_counts()`
- [ ] Build a **PivotTable** with `groupby`: one or two row fields, any summary (mean, median, count, sum)
- [ ] Build the same with `pivot_table`, rows down, columns across, grand totals
- [ ] **XLOOKUP** a second table in with `merge`, and spot the rows it didn't find (NaN)
- [ ] Find and fill **missing values**: `.isna()`, `.fillna()`
- [ ] Cut people into **quartiles** with `qcut` and cross-tabulate them

Why this matters at work

- **The pay gap by guild, town and grade is a groupby.** Exercise 3 produces the table the Council wants, worst first.
- **Quartiles and the bonus gap** (Exercises 6 and 7) are the remaining figures the Accord (and the UK regulations) require. After today you have all six.

Four big ideas

1. groupby = PivotTable	`df.groupby("department")["annual_salary"].mean()`: row field, values, summary. Two row fields: `groupby(["a", "b"])`, a list.
2. unstack / pivot_table	Turn the second row field into columns. `pivot_table(index=, columns=, values=, aggfunc=)` is the PivotTable dialog with names.
3. merge = XLOOKUP for whole tables	`df.merge(other, on="key", how="left")`. Left keeps everyone; unmatched rows get NaN. Inner keeps only matches and silently drops the rest.
4. NaN is "no value"	A failed lookup or an empty cell. `.isna().sum()` counts them, `.fillna(x)` replaces them.

Three things that will bite you

- `groupby("a", "b")` is not two keys. `groupby(["a", "b"])` is.
- `how="inner"` loses rows without a word. Print `len()` before and after every merge.
- Rounding before grouping gives slightly different means than rounding after. Round at the end.

How today works

15 MIN
Lecture
Lars rebuilds Lady Siv's eleven PivotTables, live

60 MIN
Exercise sheet
Seven ravens: the Council's papers, table by table

15 MIN
Pages from the Grimoire, page 10
"pivot_magic.py", recovered from a locked tab

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Siv	Headcount	`value_counts()`, `normalize=True`
2	The Queen	The PivotTable	`groupby().mean()` `.agg([...])` `.idxmax()`
3	Pell	Two row fields	`groupby([a, b])`, `.unstack()`
4	The Queen	pivot_table, the real one	`pivot_table(..., margins=True)`
5	Orrin	XLOOKUP	`merge()`, `.isna()` `.fillna()`
6	The Queen	Pay quartiles	`pd.qcut()`, `pd.crosstab()`
7 ⭐	Siv	The bonus gap	all of the above, `.loc[row, col]`

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if Biscuit's breakfast can stay lost), each printing the expected output
- [ ] Dalvric's pivot_magic runs with all 200 people and the right percentages, one commit per bug
- [ ] Everything is pushed to `lesson-10` and you've opened a pull request

RUNE
Rune of Council

Next lesson: Before the Throne. Charts with matplotlib and a properly formatted Excel workbook written by pandas. The numbers are done; now they have to look like something the Queen will read.